Supplementary Figures and Tables for 'Fine-scale population structure within and among Malagasy societies'

Supplementary

Data cleaning

Only autosomal SNPs were analysed. SNPs with a missingness rate above 0.05 were removed. Individuals with a genotype call rate less than 0.9 (missingness exceeding 0.1) were removed. SNPs with minor allele frequency below 0.025 were removed. Hardy-Weinberg equilibrium p-values were calculated for each SNP within each population using the tidypopgen function qc_report_loci, which provides a Bonferroni correction given the number of populations analysed. SNPs with a p-value below 0.01 in at least one population after correction were removed. Close relatives were removed using the KING robust kinship coefficient Manichaikul et al. (2010), implemented in tidypopgen. One individual was removed from each pair with a kinship coefficient greater than 0.0884, equivalent to removing relatives up to the second degree.

f_4 ratio

To obtain a chimpanzee reference, SNPs in our dataset were lifted to PanTro3.0 (GenBank accession GCA_000001515.5) from hg19 using liftOver overchain file https://hgdownload.soe.ucsc.edu/goldenPath/hg19/liftOver/hg19ToPanTro5.over.chain.gz from UCSC Genome Browser Kent et al. (2002), using bigsnpr function snp_modifyBuild() Privé et al. (2018). The chimpanzee reference for each SNP was obtained using a custom R function to query the Ensembl API (GenBank accession GCA_000001515.5, Ensembl release 115, accessed via Ensembl API https://rest.ensembl.org/documentation/info/sequence_region on 17-12-2025). This procedure retained 77993 SNPs for f_4 ratio calculation. Standard errors were calculated using block-jackknife resampling (admixtools f_4 ratio default). Supplementary Table 5 contains full f_4 ratio output.

MALDER

MALDER was run using the default -binsize of 5 * 10-4 Morgans (describing the size of the bins that SNPs are split up into. Instead of directly calculating distance between each SNP pair, MALDER uses the distance between the bins of each SNP pair.), and the –mindis parameter (minimum genetic distance to begin LD decay curve fitting) set to 0.005 cM.

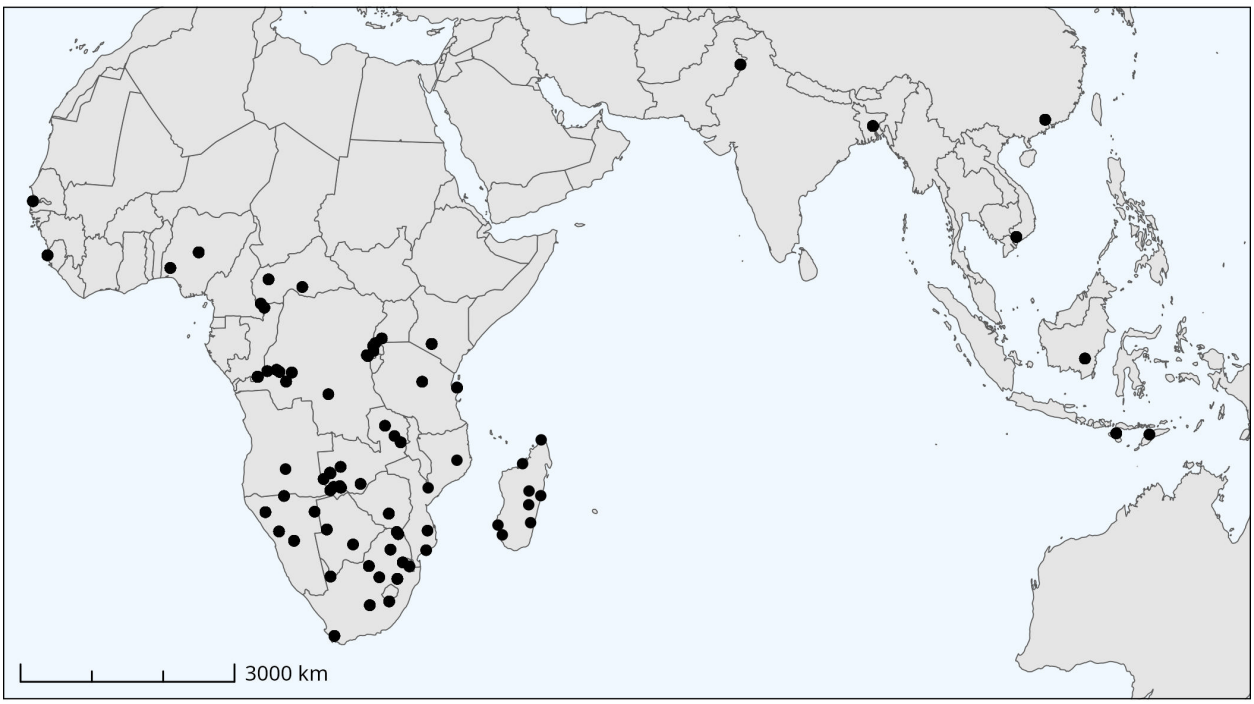


Figure 1: Map of reference population locations for all populations where coordinates were available in metadata.

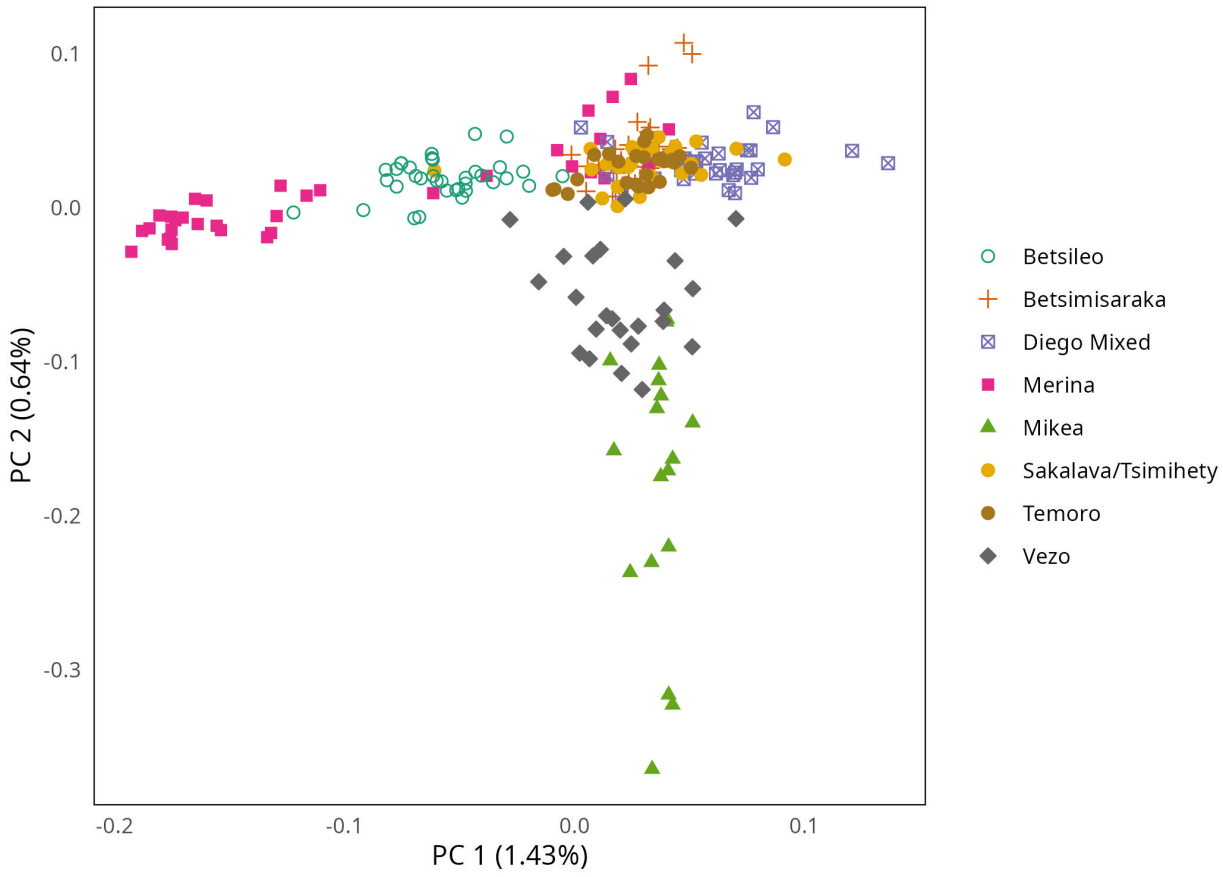


Figure 2: Principal components analysis of Malagasy data (n = 167), combined with data from Pierron et al. (2014) (Mikea, Vezo, and Temoro data points, n = 69). Colour and shape indicates self-identified ethnicity. 'Diego Mixed' individuals identified as Anjoaty, Antakarana, Sakalava Tavaratra, and Sakalava ethnicities.

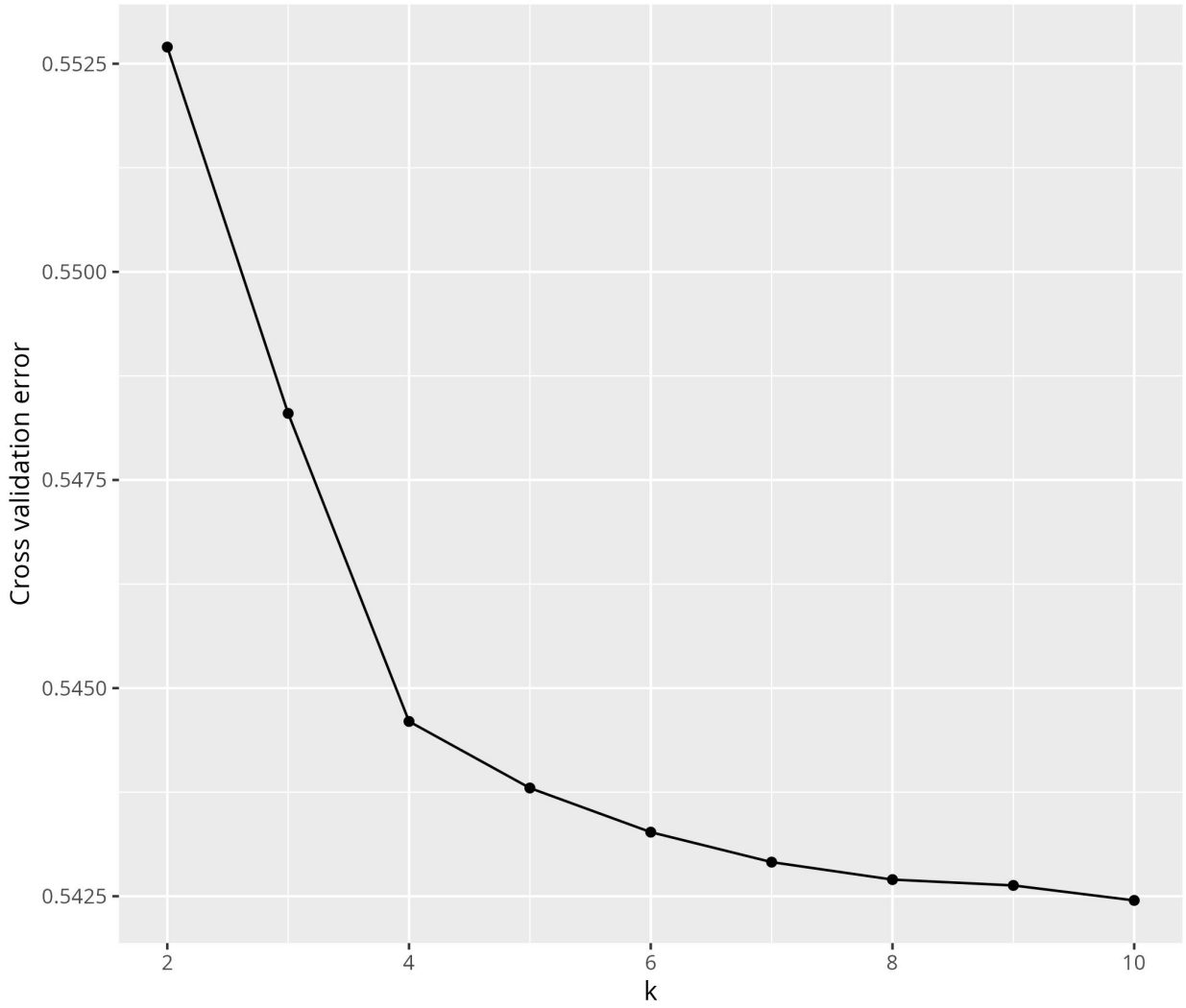


Figure 3: Cross-validation error for admixture.

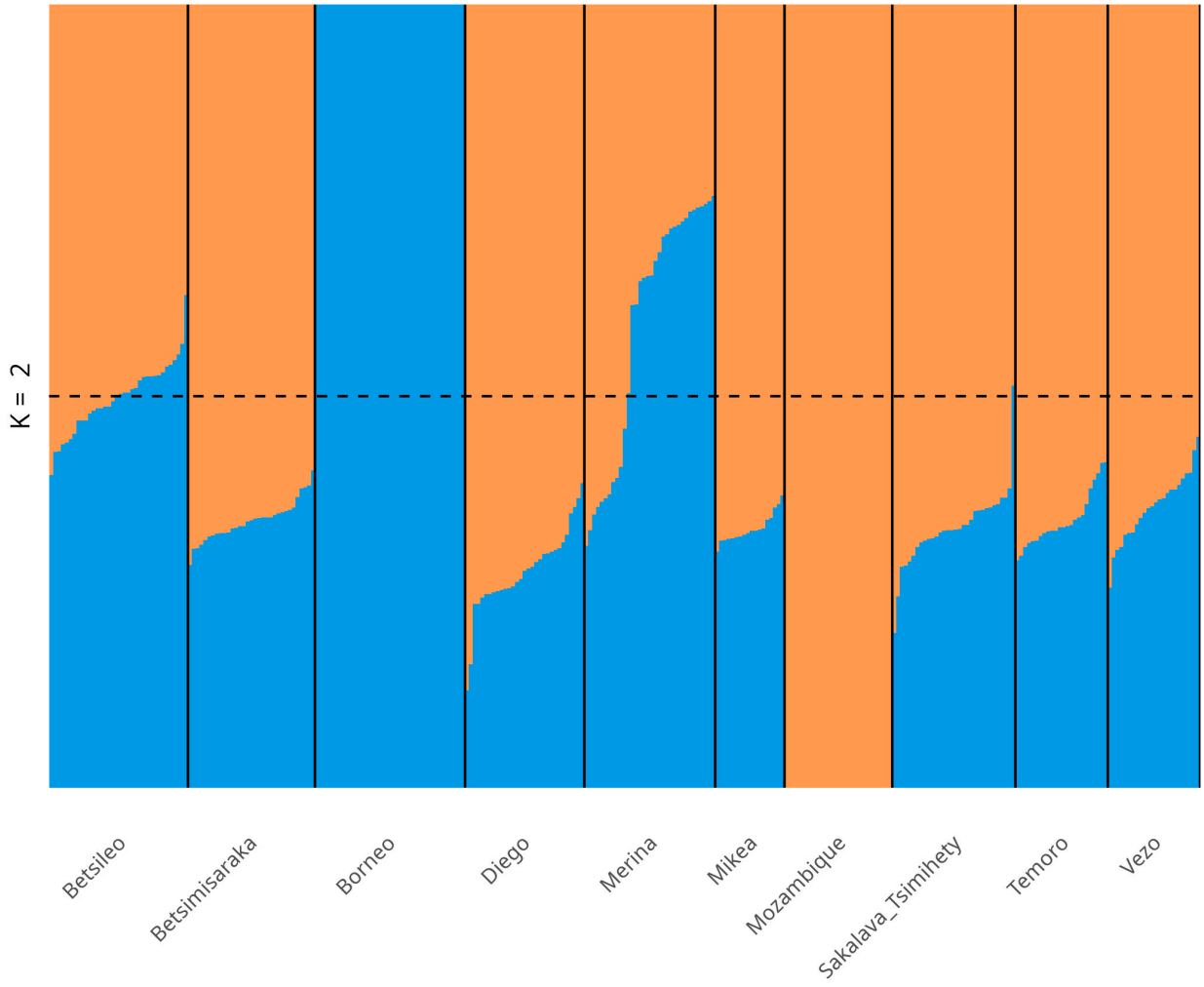


Figure 4: Supervised admixture at K = 2 for Malagasy compared to reference populations. Individuals from Borneo from Brucato et al. (2016), and individuals from Mozambique populations from Fortes-Lima et al., (2024). Mozambique individuals are labelled together, but this group contains individuals given the following population labels: “Mozambique_Nhungwe”, “Mozambique_Tsonga”, “Mozambique_Ronga”, “Mozambique_TongaMoz”, “Mozambique_Chuwabu”, “Mozambique_Tswa”, “Mozambique_Chopi”, “Mozambique_Sena”, “Mozambique_Makhuwa”.

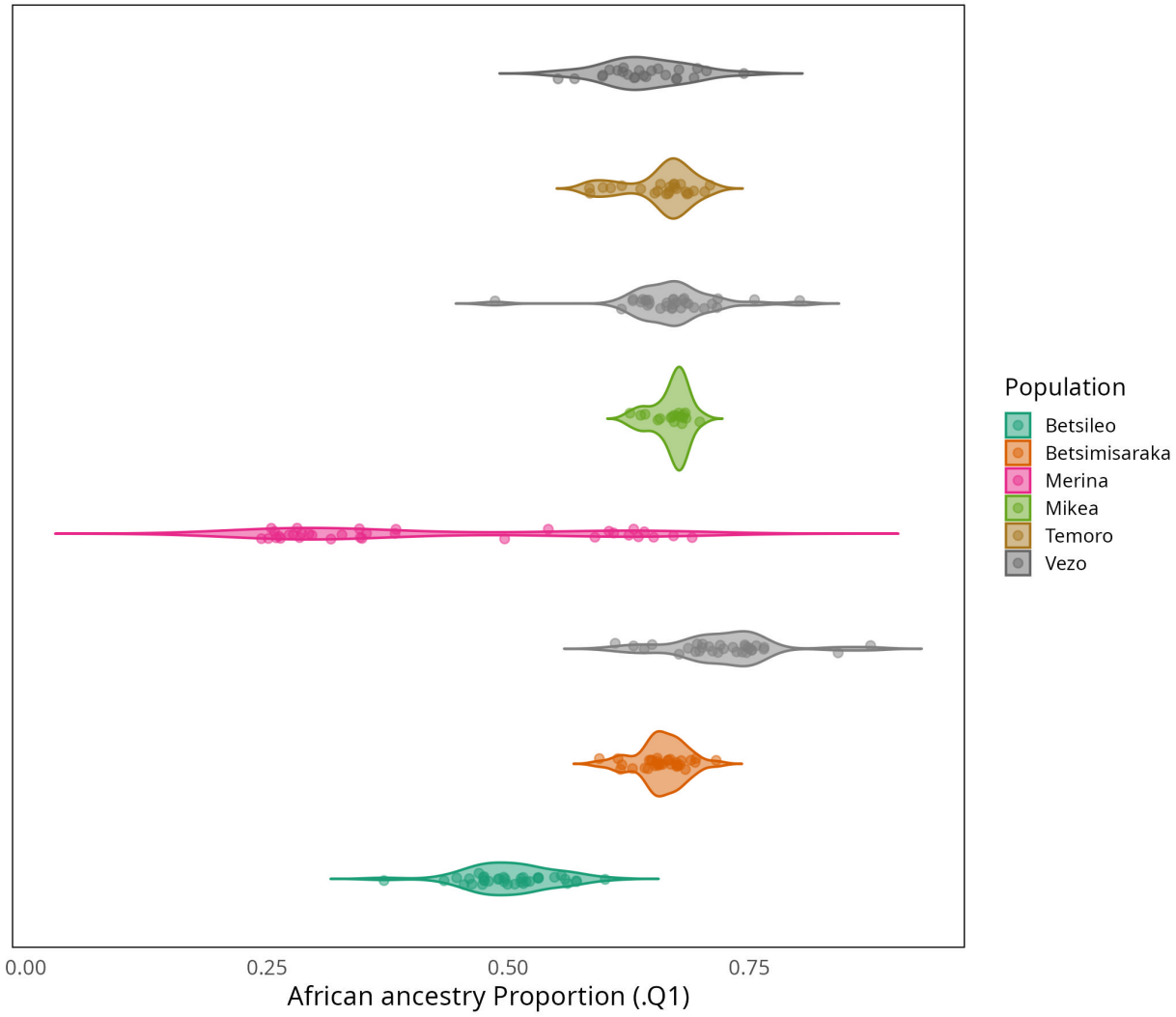


Figure 5: African ancestry proportion for each Malagasy population derived from supervised admixture at K = 2.

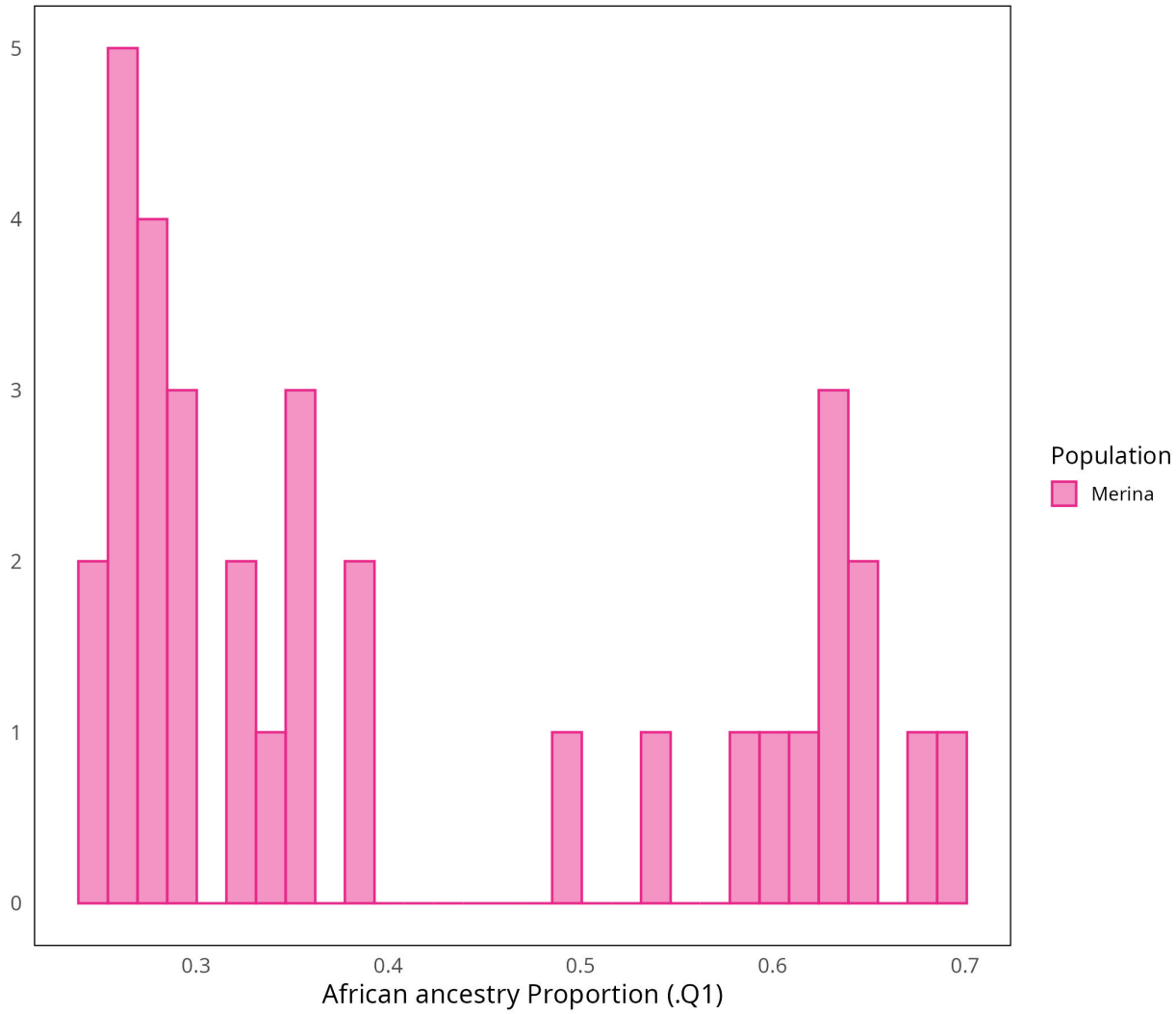


Figure 6: African ancestry proportion for Merina individuals derived from supervised admixture at K = 2, showing the bimodal distribution of ancestry among Merina individuals..

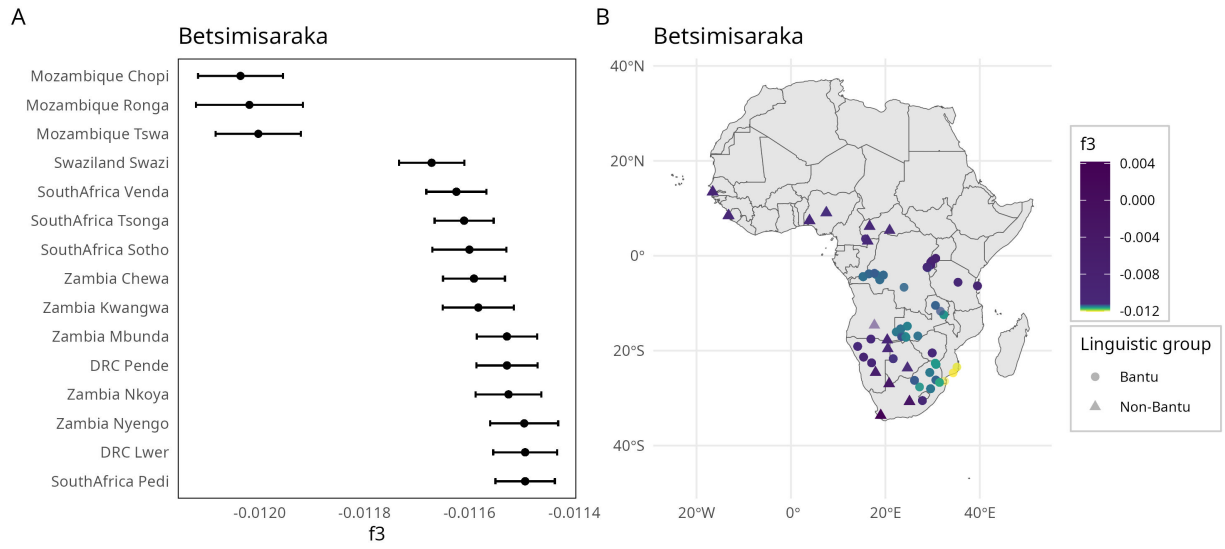


Figure 7: A) f3 statistic values across different African references for the Betsimisaraka, showing the 15 lowest f3 estimates, with Borneo as the Asian reference. B) f3 statistic values across different African references, with Borneo as the Asian reference, plotted by location for populations where coordinates were available in metadata. Colour gradient was determined by taking the standard error of the minimum f3 value and multiplying this by 10. This scale was chosen to emphasise the very best fitting f3 values, and values outside this range are coloured dark blue.

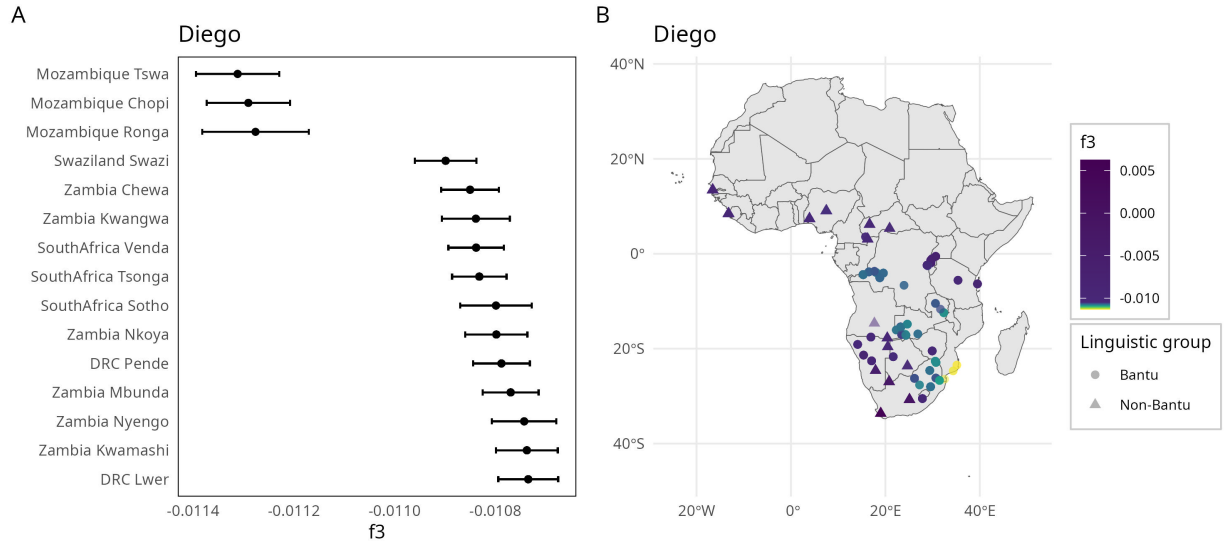


Figure 8: A) f3 statistic values across different African references for individuals from Diego, showing the 15 lowest f3 estimates, with Borneo as the Asian reference. B) f3 statistic values across different African references, with Borneo as the Asian reference, plotted by location for populations where coordinates were available in metadata. Colour gradient was determined by taking the standard error of the minimum f3 value and multiplying this by 10. This scale was chosen to emphasise the very best fitting f3 values, and values outside this range are coloured dark blue.

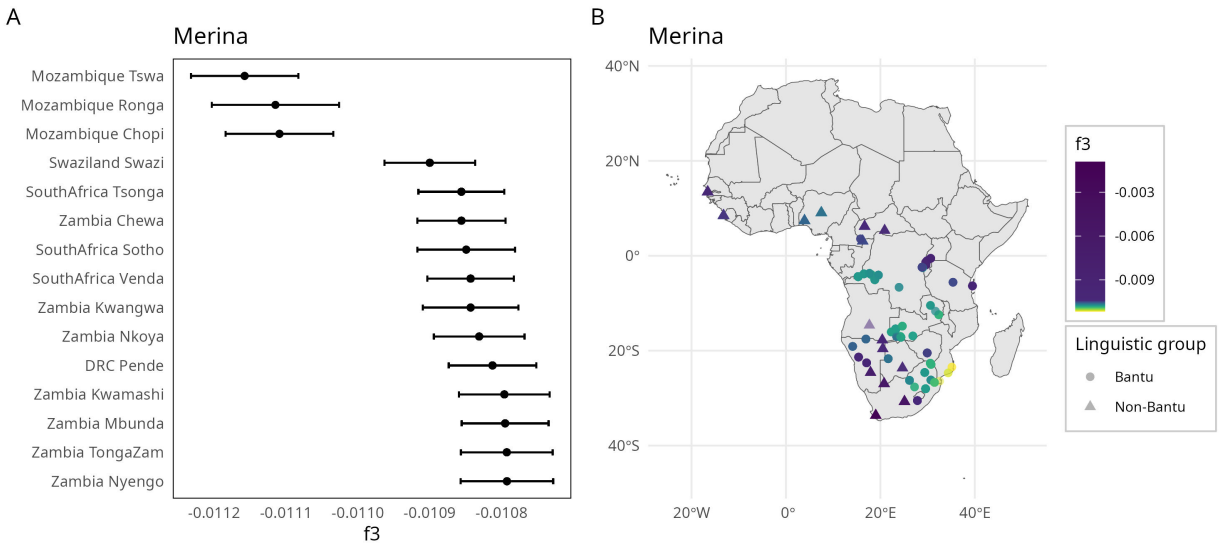


Figure 9: A) f3 statistic values across different African references for the Merina, showing the 15 lowest f3 estimates, with Borneo as the Asian reference. B) f3 statistic values across different African references, with Borneo as the Asian reference, plotted by location for populations where coordinates were available in metadata. Colour gradient was determined by taking the standard error of the minimum f3 value and multiplying this by 10. This scale was chosen to emphasise the very best fitting f3 values, and values outside this range are coloured dark blue.

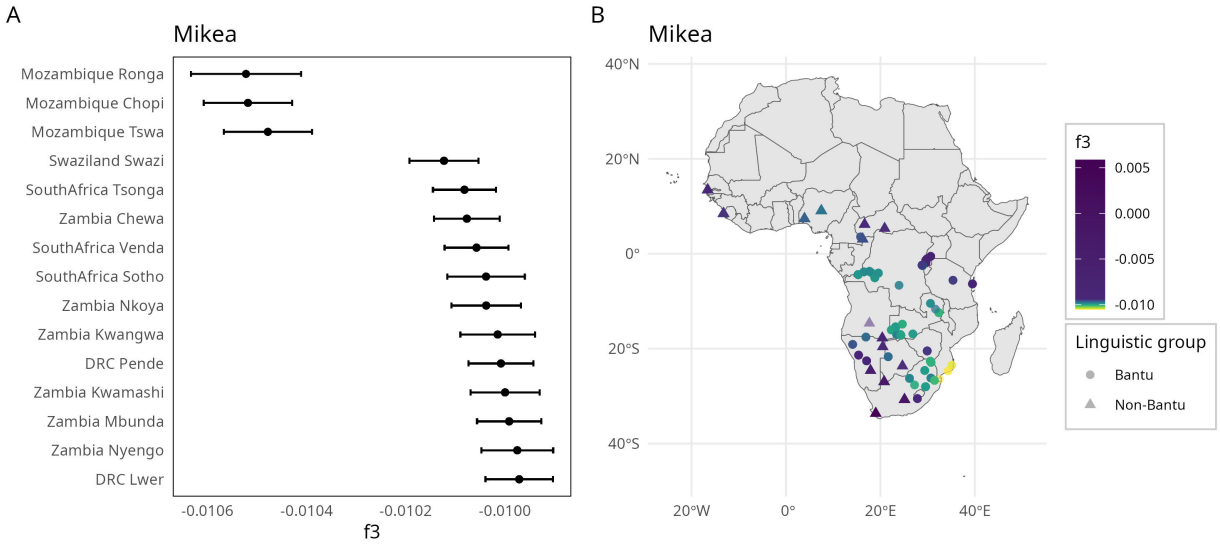


Figure 10: A) f3 statistic values across different African references for the Mikea, showing the 15 lowest f3 estimates, with Borneo as the Asian reference. B) f3 statistic values across different African references, with Borneo as the Asian reference, plotted by location for populations where coordinates were available in metadata. Colour gradient was determined by taking the standard error of the minimum f3 value and multiplying this by 10. This scale was chosen to emphasise the very best fitting f3 values, and values outside this range are coloured dark blue.

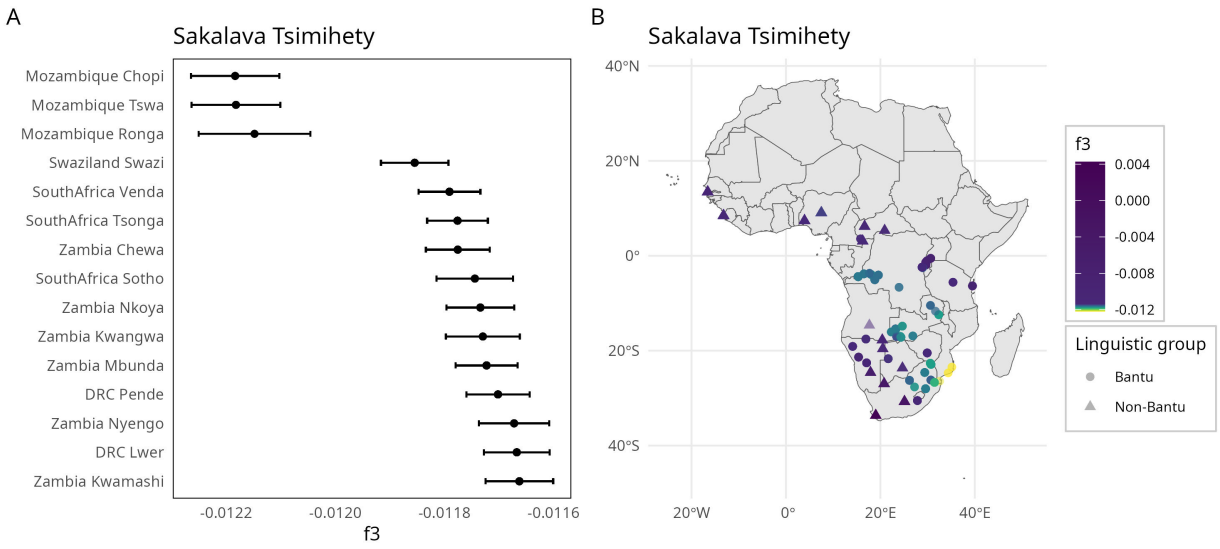


Figure 11: A) f3 statistic values across different African references for the Sakalava and Tsimihety, showing the 15 lowest f3 estimates, with Borneo as the Asian reference. B) f3 statistic values across different African references, with Borneo as the Asian reference, plotted by location for populations where coordinates were available in metadata. Colour gradient was determined by taking the standard error of the minimum f3 value and multiplying this by 10. This scale was chosen to emphasise the very best fitting f3 values, and values outside this range are coloured dark blue.

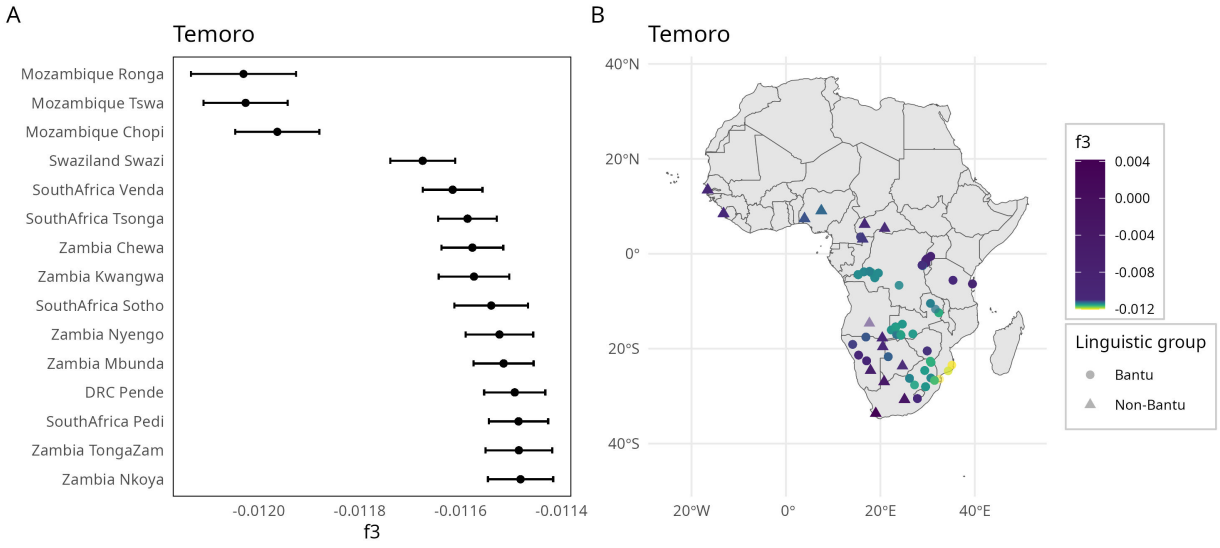


Figure 12: A) f3 statistic values across different African references for the Temoro, showing the 15 lowest f3 estimates, with Borneo as the Asian reference. B) f3 statistic values across different African references, with Borneo as the Asian reference, plotted by location for populations where coordinates were available in metadata. Colour gradient was determined by taking the standard error of the minimum f3 value and multiplying this by 10. This scale was chosen to emphasise the very best fitting f3 values, and values outside this range are coloured dark blue.

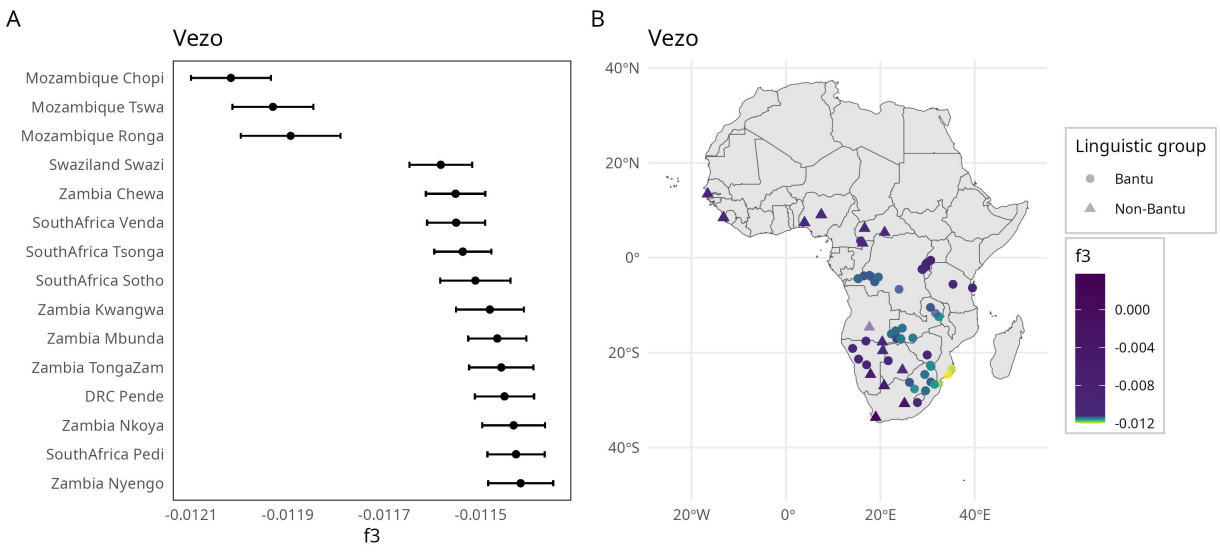


Figure 13: A) f3 statistic values across different African references for the Vezo, showing the 15 lowest f3 estimates, with Borneo as the Asian reference. B) f3 statistic values across different African references, with Borneo as the Asian reference, plotted by location for populations where coordinates were available in metadata. Colour gradient was determined by taking the standard error of the minimum f3 value and multiplying this by 10. This scale was chosen to emphasise the very best fitting f3 values, and values outside this range are coloured dark blue.

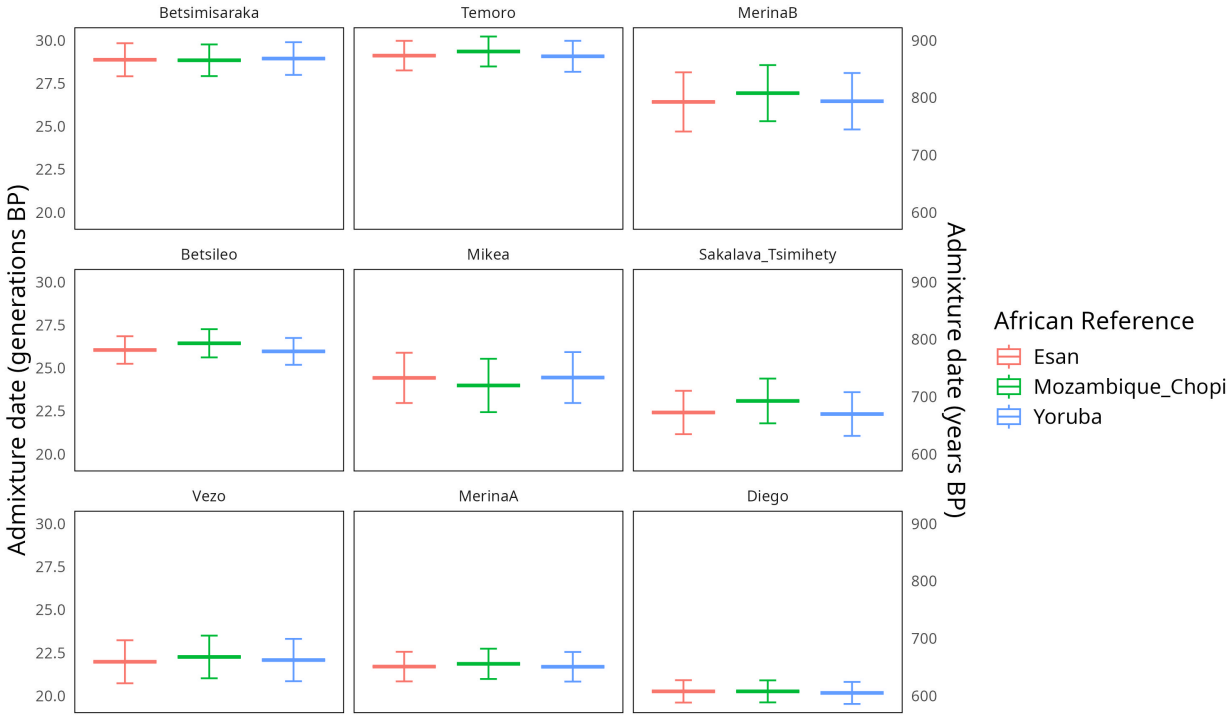


Figure 14: Estimated admixture dates for Malagasy populations using MALDER (Pickrell et al., 2014), with Borneo as an Asian source population and varied African source populations. Horizontal bar indicates estimated date of admixture, vertical bars represent standard error from leave-one-chromosome-out jackknifing.

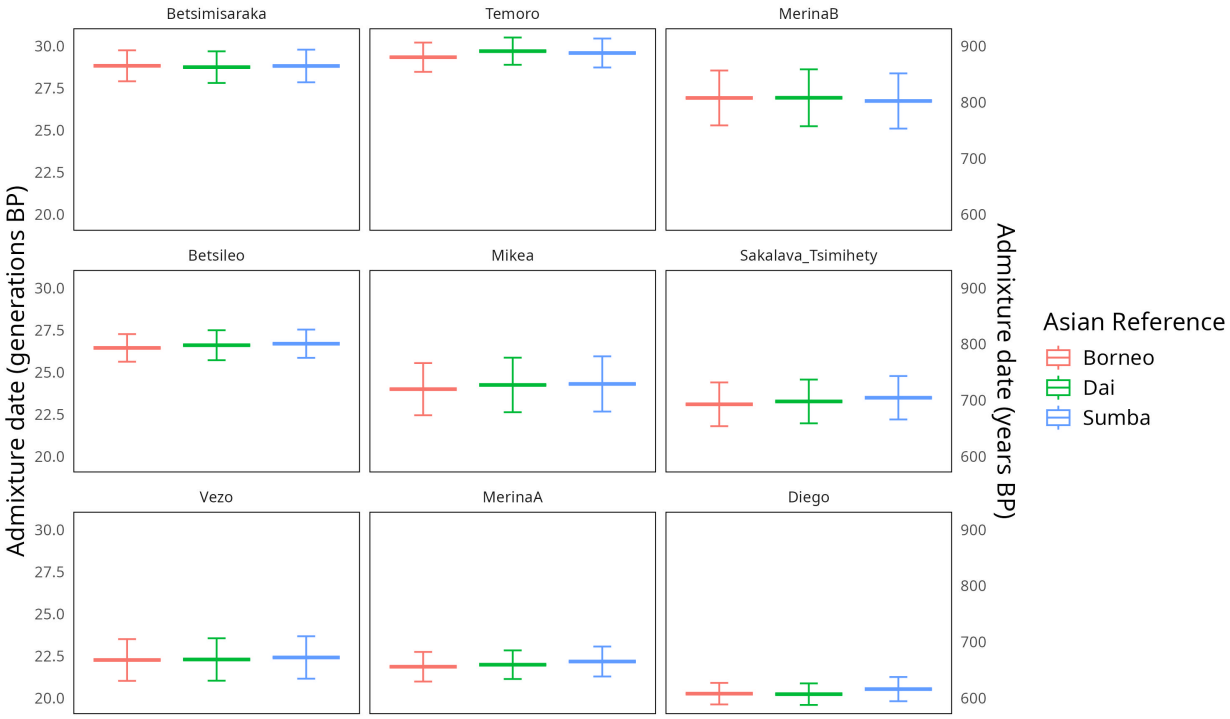


Figure 15: Estimated admixture dates for Malagasy populations using MALDER (Pickrell et al., 2014), with Mozambique as an African source population and varied Asian source populations. Horizontal bar indicates estimated date of admixture, vertical bars represent standard error from leave-one-chromosome-out jackknifing.

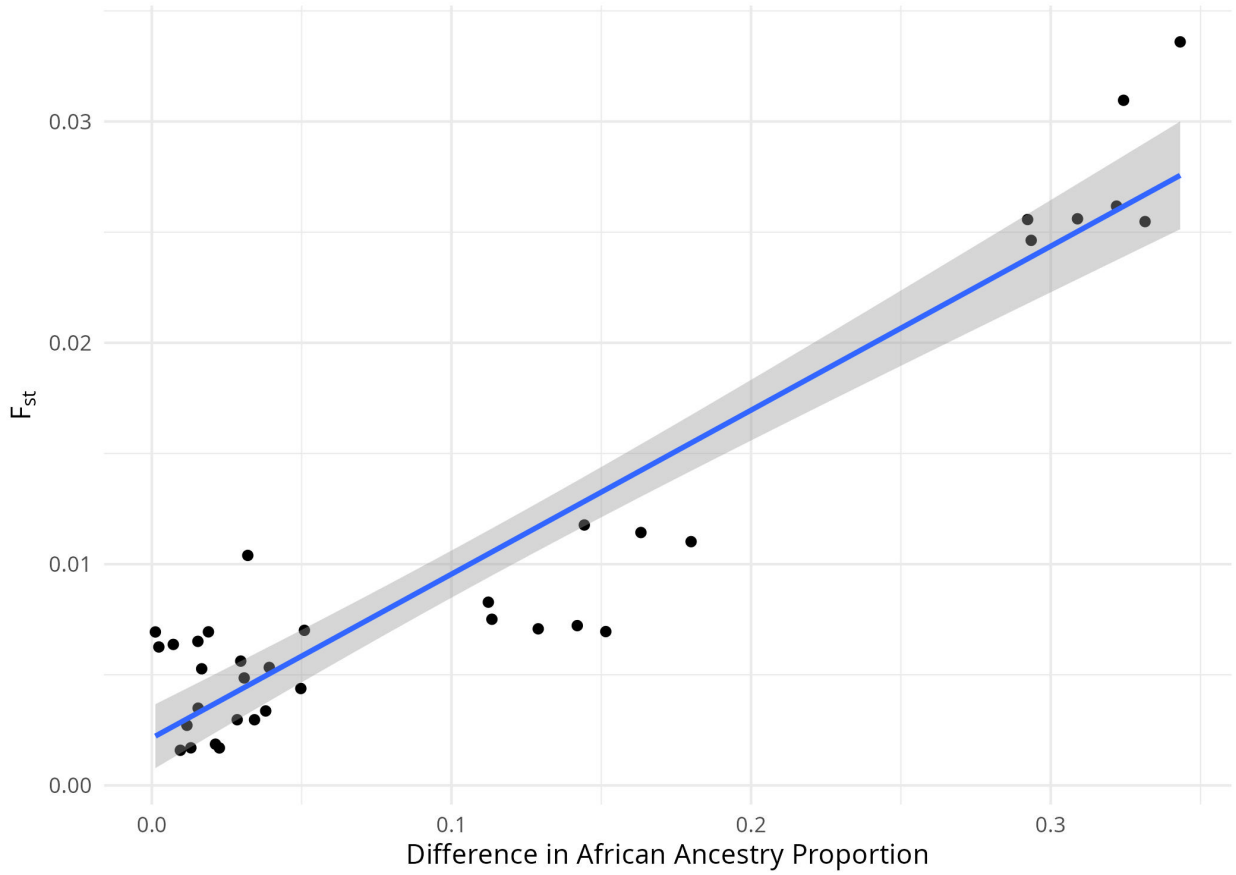


Figure 16: Difference in African ancestry proportion, as determined by f4 ratio, and Pairwise FST for pairs of Malagasy populations.

Table 1: Sample sizes for each population in the global dataset after filtering.

Population	Sample size	source
Angola_Khwe	14	Fortes-Lima et al., 2024
Angola_Sekele	7	Fortes-Lima et al., 2024
Angola_Vasekele	4	Fortes-Lima et al., 2024
Angola_Xun	2	Fortes-Lima et al., 2024
Bengali, Bengal	86	1000 Genomes Project Consortium, 2015
Betsileo	36	This study
Betsimisaraka	33	This study
Borneo	39	Brucato et al., 2016
Botswana_Ghanzi	25	Fortes-Lima et al., 2024
Botswana_GuiGhanaKgal	9	Fortes-Lima et al., 2024
Botswana_Kalanga	3	Fortes-Lima et al., 2024
Botswana_Kgalagadi	6	Fortes-Lima et al., 2024
Botswana_TswanaBot	6	Fortes-Lima et al., 2024
CAR_Banda	10	Fortes-Lima et al., 2024
CAR_DzangaShangaPeople	16	Fortes-Lima et al., 2024
CAR_Gbaya	21	Fortes-Lima et al., 2024
CAR_Mpiemo	9	Fortes-Lima et al., 2024
CAR_Nzakara	8	Fortes-Lima et al., 2024
CAR_Zande	6	Fortes-Lima et al., 2024
Han, China	105	1000 Genomes Project Consortium, 2015
Diego	31	This study
DRC_Balobo	1	Fortes-Lima et al., 2024
DRC_Bandundu	1	Fortes-Lima et al., 2024
DRC_Bangubangu	1	Fortes-Lima et al., 2024
DRC_BanguBangu	2	Fortes-Lima et al., 2024
DRC_Binji	1	Fortes-Lima et al., 2024
DRC_Binza	1	Fortes-Lima et al., 2024
DRC_Boo	1	Fortes-Lima et al., 2024
DRC_Ciokwe	1	Fortes-Lima et al., 2024
DRC_Cokwe	1	Fortes-Lima et al., 2024
DRC_Ding	42	Fortes-Lima et al., 2024
DRC_Fang	1	Fortes-Lima et al., 2024
DRC_Haavu	1	Fortes-Lima et al., 2024
DRC_Hendo	1	Fortes-Lima et al., 2024
DRC_Hungan	3	Fortes-Lima et al., 2024
DRC_Kango	1	Fortes-Lima et al., 2024
DRC_Kanyoka	1	Fortes-Lima et al., 2024
DRC_Kuba	3	Fortes-Lima et al., 2024
DRC_Kusu	1	Fortes-Lima et al., 2024
DRC_Lele	5	Fortes-Lima et al., 2024
DRC_Lika	1	Fortes-Lima et al., 2024
DRC_Luba	1	Fortes-Lima et al., 2024
DRC_LubaKasai	2	Fortes-Lima et al., 2024
DRC_LubaKatanga	2	Fortes-Lima et al., 2024
DRC_LubaLulua	6	Fortes-Lima et al., 2024

Table 1: Sample sizes for each population in the global dataset after filtering. *(continued)*

Population	Sample size	source
DRC_Lulua	1	Fortes-Lima et al., 2024
DRC_Lunda	2	Fortes-Lima et al., 2024
DRC_Luozi	2	Fortes-Lima et al., 2024
DRC_Lwer	17	Fortes-Lima et al., 2024
DRC_Mangbetu	1	Fortes-Lima et al., 2024
DRC_Manyanga	35	Fortes-Lima et al., 2024
DRC_Mbaama	1	Fortes-Lima et al., 2024
DRC_Mbala	20	Fortes-Lima et al., 2024
DRC_Mbudza	2	Fortes-Lima et al., 2024
DRC_Mbunda	1	Fortes-Lima et al., 2024
DRC_Mbuun	31	Fortes-Lima et al., 2024
DRC_MbweneNgong	1	Fortes-Lima et al., 2024
DRC_Mongo	1	Fortes-Lima et al., 2024
DRC_MongoNkundo	8	Fortes-Lima et al., 2024
DRC_Mpe	4	Fortes-Lima et al., 2024
DRC_Mpiin	2	Fortes-Lima et al., 2024
DRC_Mpur	1	Fortes-Lima et al., 2024
DRC_Myene	1	Fortes-Lima et al., 2024
DRC_Nande	1	Fortes-Lima et al., 2024
DRC_Nandi	2	Fortes-Lima et al., 2024
DRC_Ndibu	7	Fortes-Lima et al., 2024
DRC_Ngbaka	1	Fortes-Lima et al., 2024
DRC_Ngbandi	1	Fortes-Lima et al., 2024
DRC_Ngombe	2	Fortes-Lima et al., 2024
DRC_Ngong	4	Fortes-Lima et al., 2024
DRC_Ngwi	67	Fortes-Lima et al., 2024
DRC_Nienge	1	Fortes-Lima et al., 2024
DRC_Nsambaan	1	Fortes-Lima et al., 2024
DRC_Nsong	2	Fortes-Lima et al., 2024
DRC_Ntandu	4	Fortes-Lima et al., 2024
DRC_NunuBobangi	3	Fortes-Lima et al., 2024
DRC_NunuMushie	1	Fortes-Lima et al., 2024
DRC_Nzadi	7	Fortes-Lima et al., 2024
DRC_Nzebi	1	Fortes-Lima et al., 2024
DRC_Pende	21	Fortes-Lima et al., 2024
DRC_Rega	12	Fortes-Lima et al., 2024
DRC_RundiHutu	1	Fortes-Lima et al., 2024
DRC_Ruund	1	Fortes-Lima et al., 2024
DRC_Sakata	3	Fortes-Lima et al., 2024
DRC_Sengele	1	Fortes-Lima et al., 2024
DRC_Shi	129	Fortes-Lima et al., 2024
DRC_Songe	3	Fortes-Lima et al., 2024
DRC_Teke	1	Fortes-Lima et al., 2024
DRC_Tetela	3	Fortes-Lima et al., 2024
DRC_Wonk	4	Fortes-Lima et al., 2024

Table 1: Sample sizes for each population in the global dataset after filtering. *(continued)*

Population	Sample size	source
DRC_Yaka	5	Fortes-Lima et al., 2024
DRC_Yans	27	Fortes-Lima et al., 2024
DRC_Yombe	9	Fortes-Lima et al., 2024
DRC_Zimba	1	Fortes-Lima et al., 2024
Esan, Nigeria	98	1000 Genomes Project Consortium, 2015
Mandinka, The Gambia	113	1000 Genomes Project Consortium, 2015
Kinh, Vietnam	90	1000 Genomes Project Consortium, 2015
Lesotho_SothoSouthern	7	Fortes-Lima et al., 2024
Luhye, Kenya	94	1000 Genomes Project Consortium, 2015
Merina	34	This study
Mikea	18	Pierron et al., 2014
Mozambique_Chopi	4	Fortes-Lima et al., 2024
Mozambique_Chuwabu	3	Fortes-Lima et al., 2024
Mozambique_Makhuwa	1	Fortes-Lima et al., 2024
Mozambique_Nhungwe	3	Fortes-Lima et al., 2024
Mozambique_Ronga	2	Fortes-Lima et al., 2024
Mozambique_Sena	1	Fortes-Lima et al., 2024
Mozambique_TongaMoz	3	Fortes-Lima et al., 2024
Mozambique_Tsonga	7	Fortes-Lima et al., 2024
Mozambique_Tswa	4	Fortes-Lima et al., 2024
Mende, Sierra Leone	84	1000 Genomes Project Consortium, 2015
Namibia_Damara	21	Fortes-Lima et al., 2024
Namibia_Herero	23	Fortes-Lima et al., 2024
Namibia_Himba	22	Fortes-Lima et al., 2024
Namibia_Kgalagadi	8	Fortes-Lima et al., 2024
Namibia_Nama	15	Fortes-Lima et al., 2024
Namibia_TsumkweKung	10	Fortes-Lima et al., 2024
Namibia_Wambo	22	Fortes-Lima et al., 2024
Punjabi, Pakistan	96	1000 Genomes Project Consortium, 2015
Rwanda_Nkore	18	Fortes-Lima et al., 2024
Sakalava_Tsimihety	32	This study
SouthAfrica_Bhaca	12	Fortes-Lima et al., 2024
SouthAfrica_Coloured	11	Fortes-Lima et al., 2024
SouthAfrica_ColouredColesberg	12	Fortes-Lima et al., 2024
SouthAfrica_ColouredWellington	19	Fortes-Lima et al., 2024
SouthAfrica_Hlubi	2	Fortes-Lima et al., 2024
SouthAfrica_Karretjie	9	Fortes-Lima et al., 2024
SouthAfrica_Khomani	28	Fortes-Lima et al., 2024
SouthAfrica_Mputhi	3	Fortes-Lima et al., 2024
SouthAfrica_Ndebele	8	Fortes-Lima et al., 2024
SouthAfrica_Pedi	28	Fortes-Lima et al., 2024
SouthAfrica_Shona	1	Fortes-Lima et al., 2024
SouthAfrica_Sotho	7	Fortes-Lima et al., 2024
SouthAfrica_SothoNorthern	20	Fortes-Lima et al., 2024
SouthAfrica_SothoSouthern	25	Fortes-Lima et al., 2024

Table 1: Sample sizes for each population in the global dataset after filtering. *(continued)*

Population	Sample size	source
SouthAfrica_Soto	1	Fortes-Lima et al., 2024
SouthAfrica_Swazi	5	Fortes-Lima et al., 2024
SouthAfrica_Tsonga	26	Fortes-Lima et al., 2024
SouthAfrica_Tswana	32	Fortes-Lima et al., 2024
SouthAfrica_Venda	22	Fortes-Lima et al., 2024
SouthAfrica_Xhosa	30	Fortes-Lima et al., 2024
SouthAfrica_Zulu	52	Fortes-Lima et al., 2024
Sumba	29	Cox et al., 2016
Swaziland_Swazi	14	Fortes-Lima et al., 2024
Tanzania_Mixed	1	Fortes-Lima et al., 2024
Tanzania_TanzaniaMixed	14	Fortes-Lima et al., 2024
Temoro	24	Pierron et al., 2014
Timor	7	Cox et al., 2016
Uganda_Fumbira	15	Fortes-Lima et al., 2024
Uganda_Kiga	15	Fortes-Lima et al., 2024
Uganda_Konzo	5	Fortes-Lima et al., 2024
Uganda_Nkore	18	Fortes-Lima et al., 2024
Vezo	24	Pierron et al., 2014
Yoruba, Nigeria	104	1000 Genomes Project Consortium, 2015
Zambia_Bemba	11	Fortes-Lima et al., 2024
Zambia_Chewa	17	Fortes-Lima et al., 2024
Zambia_Fwe	14	Fortes-Lima et al., 2024
Zambia_Kalanga	12	Fortes-Lima et al., 2024
Zambia_Kaonde	4	Fortes-Lima et al., 2024
Zambia_Kavango	1	Fortes-Lima et al., 2024
Zambia_Kgalagadi	10	Fortes-Lima et al., 2024
Zambia_Kwamashi	16	Fortes-Lima et al., 2024
Zambia_Kwangari	1	Fortes-Lima et al., 2024
Zambia_Kwangwa	8	Fortes-Lima et al., 2024
Zambia_Lozi	23	Fortes-Lima et al., 2024
Zambia_Luyana	5	Fortes-Lima et al., 2024
Zambia_Mbukushu	8	Fortes-Lima et al., 2024
Zambia_Mbunda	25	Fortes-Lima et al., 2024
Zambia_Ngoni	7	Fortes-Lima et al., 2024
Zambia_Nkoya	14	Fortes-Lima et al., 2024
Zambia_Nsenga	4	Fortes-Lima et al., 2024
Zambia_Nyengo	10	Fortes-Lima et al., 2024
Zambia_Shanjo	9	Fortes-Lima et al., 2024
Zambia_Shona	1	Fortes-Lima et al., 2024
Zambia_ShonaZam	4	Fortes-Lima et al., 2024
Zambia_TongaZam	10	Fortes-Lima et al., 2024
Zambia_Tswana	2	Fortes-Lima et al., 2024
Zambia_Tumbuka	4	Fortes-Lima et al., 2024
Zanzibar_Swahili	22	Fortes-Lima et al., 2024
Zimbabwe_Remba	32	Fortes-Lima et al., 2024

Table 1: Sample sizes for each population in the global dataset after filtering. *(continued)*

Population	Sample size	source
Total	2744	Total

Table 2: Summary of analyses and software used in this study.

Analysis	Package/function	Dataset	Individuals Removed	n indivs	Loci filtering	n SNPs
Malagasy PCA	tidypopgen: gt_pca_randomSVD	Malagasy dataset	Missingness > 0.1; KING kinship > 0.0884	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	233	227410
Global PCA	tidypopgen: gt_pca_randomSVD	Global dataset	Missingness > 0.1; KING kinship > 0.0884	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	2744	198196
Unsupervised clustering	tidypopgen: gt_admixture	Global dataset	Missingness > 0.1; KING kinship > 0.0884	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	2744	198196
Supervised clustering	ADMIXTURE: ADMIXTURE	Subset of Global dataset	Missingness > 0.1; KING kinship > 0.0884	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	299	198196
F3 statistics	tidypopgen: gt_extract_f2; admixtools2 (version 2.0.8) (Maier and Patterson, 2024): qp3pop	Global dataset	Missingness > 0.1; KING kinship > 0.0884	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	2744	198196
F4 ratio statistics	tidypopgen: gt_extract_f2; admixtools2 (version 2.0.8) (Maier and Patterson, 2024): qpf4ratio	Global dataset + Pan and European outgroups	Missingness > 0.1; KING kinship > 0.0884	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	2792	180431
MALDER	MALDER (version 1.0) (Pickrell et al., 2014): MALDER	Global dataset	Missingness > 0.1; KING kinship > 0.0884	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025	2744	425904
Pairwise FST	tidypopgen: pairwise_pop_fst - method = "WC84" (Weir and Cockerham, 1984)	Global dataset	Missingness > 0.1; KING kinship > 0.0884	Non-autosomal; Missingness < 0.05; HWE p > 0.01; MAF > 0.025; LD clumping	2744	198196

Table 3: Minimum f3 statistic values across all combinations of references for each Malagasy population.

Malagasy	Asian	African	est	se	z	p
Betsileo	Borneo	Mozambique_Ronga	-0.0123638	0.0000925	-133.67820	0
Betsimisaraka	Borneo	Mozambique_Chopi	-0.0120364	0.0000810	-148.58346	0
Diego	Borneo	Mozambique_Tswa	-0.0113118	0.0000828	-136.57844	0
Merina	Borneo	Mozambique_Tswa	-0.0111576	0.0000747	-149.26685	0
Mikea	Borneo	Mozambique_Ronga	-0.0105223	0.0001111	-94.68672	0
Sakalava Tsimihety	Borneo	Mozambique_Chopi	-0.0121836	0.0000811	-150.27787	0
Temoro	Borneo	Mozambique_Ronga	-0.0120305	0.0001042	-115.42556	0
Vezo	Borneo	Mozambique_Chopi	-0.0120179	0.0000829	-144.89537	0

Table 4: Minimum z-scores for f3 statistics across all combinations for each Malagasy population.

Malagasy	Asian	African	est	se	z	p
Betsileo	Borneo	DRC_Shi	-0.0115276	5.51e-05	-209.2158	0
Betsimisaraka	Borneo	DRC_Shi	-0.0110031	5.16e-05	-213.0884	0
Diego	Borneo	DRC_Shi	-0.0102260	4.85e-05	-210.7807	0
Merina	Borneo	DRC_Shi	-0.0104804	5.67e-05	-184.9265	0
Mikea	Borneo	DRC_Ngwi	-0.0099074	6.14e-05	-161.3010	0
Sakalava Tsimihety	Borneo	DRC_Shi	-0.0111734	5.08e-05	-220.1194	0
Temoro	Borneo	DRC_Shi	-0.0110101	5.37e-05	-205.2034	0
Vezo	Borneo	DRC_Shi	-0.0109646	5.46e-05	-200.6620	0

Table 5: Table 5 is provided in an accompanying Excel file.

Detailed Results (See accompanying Excel file)

Table 6: f4 ratios calculated with admixtools. f4 ratios were calculated in the configurations f4(GBR,Chimpanzee;Malagasy,Borneo) / f4(GBR,Chimpanzee;Esan,Borneo) and f4(GBR,Chimpanzee;Malagasy,Esan) / f4(GBR,Chimpanzee;Borneo,Esan)

pop1	pop2	pop3	pop4	pop5	alpha	se	z
GBR	chimp_ref	Betsimisaraka	Borneo	ESN	0.6087948	0.0056688	107.39458
GBR	chimp_ref	MerinaA	Borneo	ESN	0.2773051	0.0074150	37.39799
GBR	chimp_ref	MerinaB	Borneo	ESN	0.5695955	0.0082825	68.77100
GBR	chimp_ref	Diego	Borneo	ESN	0.6205166	0.0056107	110.59508
GBR	chimp_ref	Sakalava_Tsimihety	Borneo	ESN	0.5992745	0.0054699	109.55800
GBR	chimp_ref	Betsileo	Borneo	ESN	0.4572821	0.0059435	76.93771
GBR	chimp_ref	Temoro	Borneo	ESN	0.5862443	0.0060979	96.13881
GBR	chimp_ref	Vezo	Borneo	ESN	0.5707852	0.0062701	91.03236
GBR	chimp_ref	Mikea	Borneo	ESN	0.6016209	0.0071687	83.92276

Table 7: MALDER results for all Malagasy populations, with varied African and Asian source populations. p value: significance of model tested. test pop: Malagasy target population. ref A: Asian population. ref B: African population. 2 ref z score: z score. 2 ref decay: Date, in generations, since admixture began with standard error. 2 ref amp exp: Amplitude of the LD decay curve. amp aff: Affine term estimating population substructure.

p_value	test_pop	ref A	ref B	2-ref z-score	2-ref decay	2-ref amp_exp	amp_aff
4.6e-222	Betsileo	Borneo	Esan	31.81	26.05 +/- 0.80	0.00216033 +/- 0.00006791	0.0000299
3.6e-234	Betsileo	Borneo	Yoruba	32.67	25.97 +/- 0.78	0.00214609 +/- 0.00006568	0.0000296
5.7e-230	Betsileo	Borneo	Mozambique_Chopi	32.38	26.44 +/- 0.82	0.00227784 +/- 0.00006829	0.0000310
3.5e-188	Betsileo	Sumba	Esan	29.26	26.26 +/- 0.82	0.00191612 +/- 0.00006549	0.0000261
1.7e-196	Betsileo	Sumba	Yoruba	29.91	26.18 +/- 0.80	0.00190193 +/- 0.00006360	0.0000258
1.5e-200	Betsileo	Sumba	Mozambique_Chopi	30.22	26.69 +/- 0.84	0.00202856 +/- 0.00006714	0.0000271
8.6e-200	Betsileo	Dai	Esan	30.16	26.21 +/- 0.87	0.00216652 +/- 0.00006781	0.0000304
1.5e-205	Betsileo	Dai	Yoruba	30.59	26.14 +/- 0.85	0.00215206 +/- 0.00006551	0.0000300
1.3e-197	Betsileo	Dai	Mozambique_Chopi	29.99	26.60 +/- 0.89	0.00228532 +/- 0.00006899	0.0000315
1.1e-196	Betsimisaraka	Borneo	Esan	29.92	28.87 +/- 0.96	0.00187297 +/- 0.00004607	0.0000083
7.2e-204	Betsimisaraka	Borneo	Yoruba	30.47	28.94 +/- 0.95	0.00186972 +/- 0.00004541	0.0000081
6.9e-217	Betsimisaraka	Borneo	Mozambique_Chopi	31.43	28.84 +/- 0.92	0.00197274 +/- 0.00004997	0.0000086
4.5e-185	Betsimisaraka	Sumba	Esan	29.01	28.90 +/- 1.00	0.00165692 +/- 0.00005122	0.0000072
2.2e-189	Betsimisaraka	Sumba	Yoruba	29.35	28.96 +/- 0.99	0.00165296 +/- 0.00005075	0.0000070
1.6e-193	Betsimisaraka	Sumba	Mozambique_Chopi	29.68	28.83 +/- 0.97	0.00175133 +/- 0.00005520	0.0000074
1.3e-199	Betsimisaraka	Dai	Esan	30.14	28.81 +/- 0.96	0.00186998 +/- 0.00004787	0.0000080
6.6e-205	Betsimisaraka	Dai	Yoruba	30.54	28.88 +/- 0.95	0.00186641 +/- 0.00004697	0.0000078
3.4e-204	Betsimisaraka	Dai	Mozambique_Chopi	30.49	28.76 +/- 0.94	0.00196995 +/- 0.00005151	0.0000083
1.5e-144	Diego	Borneo	Esan	25.60	20.26 +/- 0.65	0.00146813 +/- 0.00005735	0.0000493
4.1e-148	Diego	Borneo	Yoruba	25.92	20.17 +/- 0.64	0.00145881 +/- 0.00005628	0.0000494
4.4e-159	Diego	Borneo	Mozambique_Chopi	26.87	20.26 +/- 0.64	0.00153534 +/- 0.00005713	0.0000514
2.2e-116	Diego	Sumba	Esan	22.93	20.57 +/- 0.73	0.00131657 +/- 0.00005741	0.0000432
9.8e-120	Diego	Sumba	Yoruba	23.27	20.46 +/- 0.71	0.00130741 +/- 0.00005619	0.0000433
6.7e-124	Diego	Sumba	Mozambique_Chopi	23.67	20.53 +/- 0.72	0.00137919 +/- 0.00005826	0.0000452
4.7e-130	Diego	Dai	Esan	24.26	20.24 +/- 0.66	0.00146933 +/- 0.00006055	0.0000496
1.1e-133	Diego	Dai	Yoruba	24.60	20.14 +/- 0.64	0.00145999 +/- 0.00005934	0.0000497
2.3e-138	Diego	Dai	Mozambique_Chopi	25.04	20.23 +/- 0.64	0.00153750 +/- 0.00006140	0.0000517
1.2e-228	Merina	Borneo	Esan	32.28	22.65 +/- 0.70	0.00176093 +/- 0.00005003	0.0004215
5.5e-236	Merina	Borneo	Yoruba	32.80	22.66 +/- 0.69	0.00175499 +/- 0.00004855	0.0004206
1.5e-226	Merina	Borneo	Mozambique_Chopi	32.13	23.18 +/- 0.72	0.00186187 +/- 0.00004419	0.0004401
1.1e-227	Merina	Sumba	Esan	32.21	22.68 +/- 0.65	0.00154560 +/- 0.00004798	0.0003678
2.2e-238	Merina	Sumba	Yoruba	32.97	22.68 +/- 0.64	0.00153947 +/- 0.00004669	0.0003670
1.6e-250	Merina	Sumba	Mozambique_Chopi	33.81	23.32 +/- 0.69	0.00164251 +/- 0.00004452	0.0003852
3.8e-233	Merina	Dai	Esan	32.60	22.74 +/- 0.70	0.00176475 +/- 0.00005140	0.0004208

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p_value	test_pop	ref A	ref B	2-ref z-score	2-ref decay	2-ref amp_exp	amp_aff
3.9e-244	Merina	Dai	Yoruba	33.37	22.75 +/- 0.68	0.00175932 +/- 0.00004984	0.0004199
1.8e-241	Merina	Dai	Mozambique_Chopi	33.19	23.29 +/- 0.70	0.00186752 +/- 0.00004541	0.0004395
5.5e-141	MerinaA	Borneo	Esan	25.28	21.70 +/- 0.86	0.00172501 +/- 0.00006674	0.0000510
2.5e-139	MerinaA	Borneo	Yoruba	25.13	21.69 +/- 0.86	0.00172051 +/- 0.00006620	0.0000510
1.8e-135	MerinaA	Borneo	Mozambique_Chopi	24.77	21.86 +/- 0.88	0.00180736 +/- 0.00006513	0.0000546
1.8e-128	MerinaA	Sumba	Esan	24.11	21.96 +/- 0.86	0.00151669 +/- 0.00006290	0.0000432
4.2e-129	MerinaA	Sumba	Yoruba	24.17	21.94 +/- 0.86	0.00151185 +/- 0.00006254	0.0000433
8.1e-137	MerinaA	Sumba	Mozambique_Chopi	24.90	22.17 +/- 0.89	0.00159416 +/- 0.00006327	0.0000466
1.4e-149	MerinaA	Dai	Esan	26.05	21.80 +/- 0.84	0.00173212 +/- 0.00006550	0.0000519
1.3e-149	MerinaA	Dai	Yoruba	26.05	21.80 +/- 0.84	0.00172795 +/- 0.00006479	0.0000519
1.2e-146	MerinaA	Dai	Mozambique_Chopi	25.79	21.98 +/- 0.85	0.00181659 +/- 0.00006355	0.0000556
2.3e-53	MerinaB	Borneo	Esan	15.38	26.42 +/- 1.72	0.00182359 +/- 0.00006434	0.0000235
1.2e-58	MerinaB	Borneo	Yoruba	16.15	26.46 +/- 1.64	0.00181557 +/- 0.00006173	0.0000232
4.2e-61	MerinaB	Borneo	Mozambique_Chopi	16.49	26.93 +/- 1.63	0.00193439 +/- 0.00005846	0.0000234
5.1e-52	MerinaB	Sumba	Esan	15.18	26.11 +/- 1.72	0.00159222 +/- 0.00006515	0.0000201
8.6e-57	MerinaB	Sumba	Yoruba	15.88	26.13 +/- 1.65	0.00158453 +/- 0.00006284	0.0000198
8.8e-60	MerinaB	Sumba	Mozambique_Chopi	16.31	26.75 +/- 1.64	0.00170122 +/- 0.00005956	0.0000200
1.9e-49	MerinaB	Dai	Esan	14.78	26.44 +/- 1.79	0.00181330 +/- 0.00006328	0.0000236
4.9e-54	MerinaB	Dai	Yoruba	15.48	26.48 +/- 1.71	0.00180611 +/- 0.00006077	0.0000233
3.8e-57	MerinaB	Dai	Mozambique_Chopi	15.93	26.94 +/- 1.69	0.00192369 +/- 0.00005553	0.0000236
6.2e-63	Mikea	Borneo	Esan	16.74	24.43 +/- 1.46	0.00179737 +/- 0.00007812	0.0000009
4.2e-61	Mikea	Borneo	Yoruba	16.49	24.45 +/- 1.48	0.00179523 +/- 0.00007745	0.0000008
8.6e-54	Mikea	Borneo	Mozambique_Chopi	15.44	23.99 +/- 1.55	0.00184531 +/- 0.00007972	0.0000014
6.7e-59	Mikea	Sumba	Esan	16.18	24.75 +/- 1.53	0.00158879 +/- 0.00007863	0.0000010
1.3e-56	Mikea	Sumba	Yoruba	15.86	24.77 +/- 1.56	0.00158630 +/- 0.00007822	0.0000009
6.3e-50	Mikea	Sumba	Mozambique_Chopi	14.86	24.30 +/- 1.64	0.00163606 +/- 0.00008013	0.0000015
2.2e-58	Mikea	Dai	Esan	16.11	24.68 +/- 1.53	0.00181233 +/- 0.00008078	0.0000010
1.1e-56	Mikea	Dai	Yoruba	15.87	24.70 +/- 1.56	0.00181012 +/- 0.00008012	0.0000009
2.4e-50	Mikea	Dai	Mozambique_Chopi	14.92	24.24 +/- 1.62	0.00186109 +/- 0.00008254	0.0000015
1.1e-70	Sakalava_Tsimihety	Borneo	Esan	17.77	22.42 +/- 1.26	0.00177727 +/- 0.00007455	0.0000396
4.3e-69	Sakalava_Tsimihety	Borneo	Yoruba	17.57	22.33 +/- 1.27	0.00176631 +/- 0.00007391	0.0000395
3.3e-70	Sakalava_Tsimihety	Borneo	Mozambique_Chopi	17.71	23.09 +/- 1.30	0.00188537 +/- 0.00007484	0.0000422
1.4e-73	Sakalava_Tsimihety	Sumba	Esan	18.14	22.70 +/- 1.25	0.00157199 +/- 0.00007657	0.0000344
3.5e-71	Sakalava_Tsimihety	Sumba	Yoruba	17.84	22.60 +/- 1.27	0.00156087 +/- 0.00007598	0.0000343

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p_value	test_pop	ref A	ref B	2-ref z-score	2-ref decay	2-ref amp_exp	amp_aff
1.2e-73	Sakalava_Tsimihety	Sumba	Mozambique_Chopi	18.15	23.48 +/- 1.29	0.00167853 +/- 0.00007755	0.0000367
1.4e-72	Sakalava_Tsimihety	Dai	Esan	18.02	22.60 +/- 1.25	0.00177662 +/- 0.00007776	0.0000398
3.9e-71	Sakalava_Tsimihety	Dai	Yoruba	17.83	22.51 +/- 1.26	0.00176524 +/- 0.00007692	0.0000396
5.3e-72	Sakalava_Tsimihety	Dai	Mozambique_Chopi	17.94	23.26 +/- 1.30	0.00188434 +/- 0.00007879	0.0000423
9.6e-133	Temoro	Borneo	Esan	24.52	29.11 +/- 0.86	0.00194337 +/- 0.00007926	0.0000182
1e-128	Temoro	Borneo	Yoruba	24.14	29.07 +/- 0.90	0.00193347 +/- 0.00008011	0.0000178
7.4e-147	Temoro	Borneo	Mozambique_Chopi	25.81	29.35 +/- 0.87	0.00205756 +/- 0.00007973	0.0000186
5.3e-104	Temoro	Sumba	Esan	21.66	29.34 +/- 0.87	0.00172607 +/- 0.00007970	0.0000165
2.7e-101	Temoro	Sumba	Yoruba	21.37	29.28 +/- 0.90	0.00171615 +/- 0.00008032	0.0000162
7.5e-112	Temoro	Sumba	Mozambique_Chopi	22.47	29.60 +/- 0.86	0.00183343 +/- 0.00008158	0.0000169
5.4e-130	Temoro	Dai	Esan	24.26	29.48 +/- 0.82	0.00195981 +/- 0.00008079	0.0000181
6.1e-127	Temoro	Dai	Yoruba	23.97	29.42 +/- 0.85	0.00194883 +/- 0.00008131	0.0000178
2.8e-142	Temoro	Dai	Mozambique_Chopi	25.40	29.71 +/- 0.81	0.00207391 +/- 0.00008166	0.0000185
1.9e-69	Vezo	Borneo	Esan	17.62	21.98 +/- 1.25	0.00184944 +/- 0.00006413	0.0000285
5.5e-72	Vezo	Borneo	Yoruba	17.94	22.08 +/- 1.23	0.00184557 +/- 0.00006245	0.0000284
1.2e-72	Vezo	Borneo	Mozambique_Chopi	18.03	22.26 +/- 1.24	0.00194431 +/- 0.00005711	0.0000301
2.4e-67	Vezo	Sumba	Esan	17.34	22.16 +/- 1.28	0.00163475 +/- 0.00005995	0.0000246
1.6e-70	Vezo	Sumba	Yoruba	17.76	22.26 +/- 1.25	0.00163029 +/- 0.00005825	0.0000245
1.3e-70	Vezo	Sumba	Mozambique_Chopi	17.76	22.41 +/- 1.26	0.00172420 +/- 0.00005415	0.0000261
4.2e-66	Vezo	Dai	Esan	17.17	22.04 +/- 1.28	0.00185290 +/- 0.00006451	0.0000280
1.1e-68	Vezo	Dai	Yoruba	17.51	22.13 +/- 1.26	0.00184873 +/- 0.00006279	0.0000278
3.5e-70	Vezo	Dai	Mozambique_Chopi	17.71	22.29 +/- 1.26	0.00194813 +/- 0.00005747	0.0000296

Table 8: Pairwise FST values between all Malagasy populations.

	Betsileo	Betsimisaraka	Diego.Mixed	Merina.A	Merina.B	Mikea	Sakalava.Tsimihety	Temoro	Vezo
Betsileo	NA	0.0069522	0.0114288	0.0110161	0.0082805	0.0117686	0.0072215	0.0070768	0.0075091
Betsimisaraka	0.0069522	NA	0.0027162	0.0254780	0.0053308	0.0063718	0.0015886	0.0016943	0.0033650
Diego Mixed	0.0114288	0.0027162	NA	0.0335991	0.0070089	0.0069401	0.0018644	0.0029681	0.0043794
Merina A	0.0110161	0.0254780	0.0335991	NA	0.0255723	0.0309598	0.0261779	0.0256050	0.0246308
Merina B	0.0082805	0.0053308	0.0070089	0.0255723	NA	0.0103931	0.0056181	0.0052702	0.0069354
Mikea	0.0117686	0.0063718	0.0069401	0.0309598	0.0103931	NA	0.0062564	0.0065102	0.0048565
Sakalava/Tsimihety	0.0072215	0.0015886	0.0018644	0.0261779	0.0056181	0.0062564	NA	0.0016993	0.0029689
Temoro	0.0070768	0.0016943	0.0029681	0.0256050	0.0052702	0.0065102	0.0016993	NA	0.0034909
Vezo	0.0075091	0.0033650	0.0043794	0.0246308	0.0069354	0.0048565	0.0029689	0.0034909	NA

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